

LAB 03

AIM: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

1. Identify all entity types and entity sets.

Researcher-Represents all researchers working in the university.

Department-Represents all departments in the university.

Research Project-Represents all research projects undertaken by the university.

Student-Represents all students working under researchers.

Publication-Represents all publications produced from research projects.

Project Assistant-Represents all assistants associated with research projects.

2. Identify strong and weak entity sets.

Strong and Weak Entity Sets

- **Strong Entity Sets:**

- Researcher
- Department
- Research Project
- Student
- Publication

- **Weak Entity Set:**

- Project Assistant

3. Identify the owner entity and partial key of the weak entity.

Weak Entity: Project Assistant

Owner Entity: Research Project

Partial Key: Assistant_Name

4. Identify simple attributes.

The simple attributes in the University Research Management System are:

- **Researcher:** Researcher_ID, Date_of_Birth, Email
- **Department:** Department_ID, Department_Name, Location
- **Research Project:** Project_ID, Project_Title, Start_Date, End_Date, Budget
- **Student:** Student_ID, Name, Email, Date_of_Birth
- **Publication:** Publication_ID, Title, Publication_Date, DOI, Publisher
- **Project Assistant:** Assistant_Name, Role, Joining_Date

5. Identify composite attributes.

The composite attributes in the system are:

1. **Researcher** → **Name**

- First_Name
- Middle_Name
- Last_Name

2. **Researcher** → **Address**

- Street
- City
- State
- PIN_Code

6. Identify single-valued attributes.

Researcher

- Researcher_ID
- Name (composite, but single-valued)
- Date_of_Birth
- Email
- Address (composite, but single-valued)

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget

Student

- Student_ID
- Name
- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

Project Assistant

- Assistant_Name
- Role
- Joining_Date

7. Identify multi-valued attributes

The multi-valued attributes in the system are:

1. **Researcher → Phone_Number**
A researcher can have multiple phone numbers.
2. **Research Project → Keyword**
A research project can have multiple keywords.

8. Identify stored attributes.

Researcher

- Researcher_ID
- Name (First_Name, Middle_Name, Last_Name)
- Date_of_Birth
- Email
- Phone_Number
- Address (Street, City, State, PIN_Code)

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget
- Keyword

Student

- Student_ID
- Name
- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

Project Assistant

- Assistant_Name
- Role
- Joining_Date

9. Identify derived attributes.

Derived Attribute- Age

Age is derived from Date_of_Birth.

10. Identify key attributes.

Entity	Key Attribute	Key Status
Researcher	Researcher_ID	Primary Key
Department	Department_ID	Primary Key
Research Project	Project_ID	Primary Key
Student	Student_ID	Primary Key
Publication	Publication_ID	Primary Key
Project Assistant	Assistant_Name	Partial Key
Project Assistant	Project_ID + Assistant_Name	Composite Key

11. Identify candidate keys and select appropriate primary keys.

Entity	Candidate Key	Selected Primary Key
Researcher	Researcher_ID	Researcher_ID
Department	Department_ID	Department_ID
Research Project	Project_ID	Project_ID
Student	Student_ID	Student_ID
Publication	Publication_ID	Publication_ID
Project Assistant	(Project_ID, Assistant_Name)	(Project_ID, Assistant_Name)

12. Identify at least one composite key.

Assitant_Name

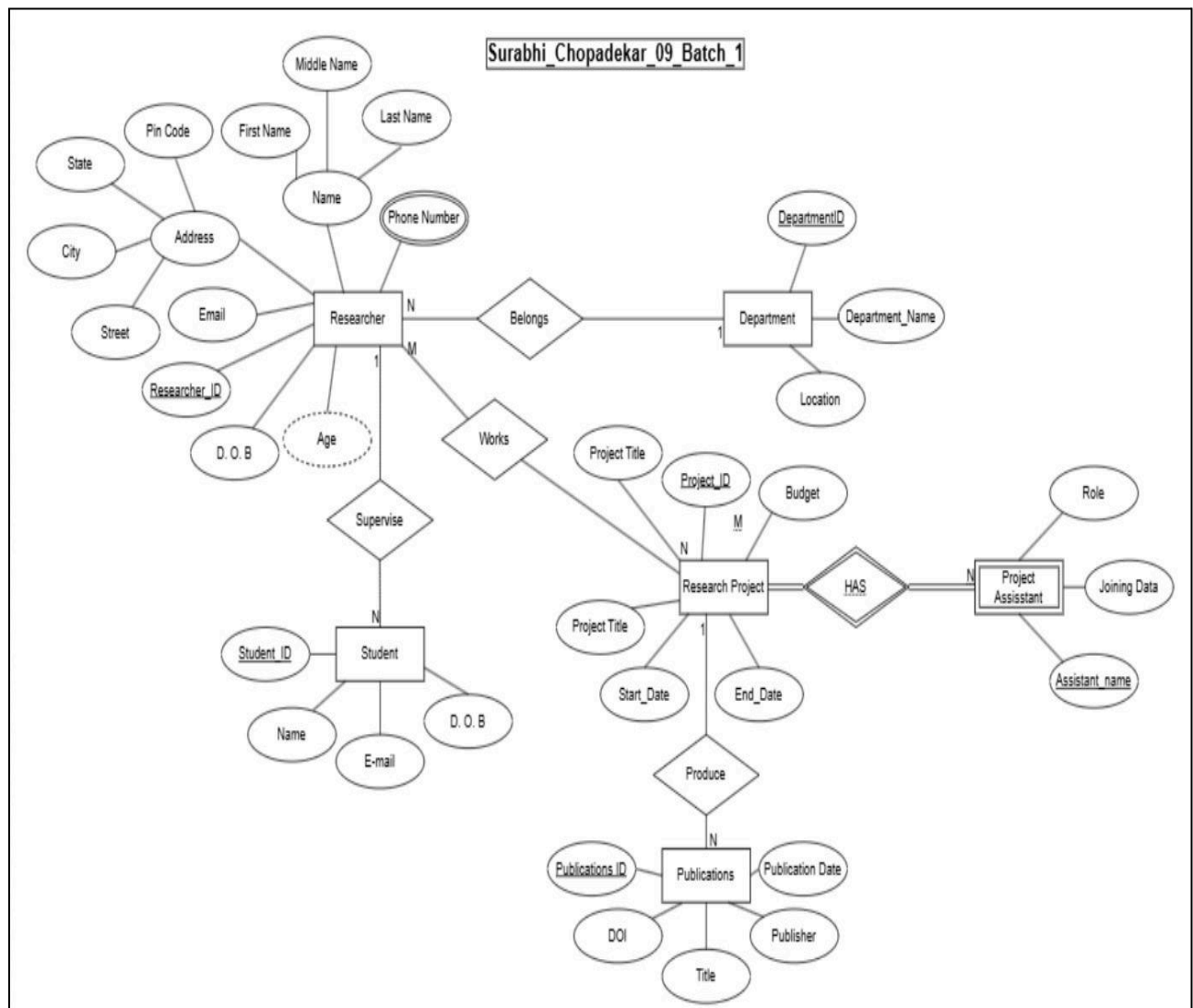
13. Identify appropriate foreign keys after converting the ER model into relations.

Relation	Foreign Key	References
RESEARCHER	Department ID	DEPARTMENT(Department ID)
RESEARCHER PROJECT	Researcher ID	RESEARCHER(Researcher ID)
RESEARCHER PROJECT	Project ID	RESEARCH PROJECT(Project ID)
RESEARCHER STUDENT	Researcher ID	RESEARCHER(Researcher ID)
RESEARCHER STUDENT	Student ID	STUDENT(Student ID)
PUBLICATION	Project ID	RESEARCH PROJECT(Project ID)
PROJECT ASSISTANT	Project ID	RESEARCH PROJECT(Project ID)

14. Identify all major relationships between the entities.

Relationship	Entities Involved	Cardinality	Meaning
BELONGS	Department ↔ Researcher	1 : N	One department has many researchers; each researcher belongs to one department.
WORKS	Researcher ↔ Research Project	M : N	A researcher can work on many projects, and a project can involve many researchers.
SUPERVISES	Researcher ↔ Student	1 : N	One researcher can supervise many students; each student works under one researcher.
PRODUCES	Research Project ↔ Publication	1 : N	One project can produce many publications; each publication belongs to one project.
HAS_ASSISTANT	Research Project ↔ Project Assistant	1 : N	One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity.

15. Draw the complete ER diagram using standard ER notation.



16. Prepare a table showing each attribute and its attribute type and key status

Entity	Attribute	Attribute Type	Value	Key Status
Researcher	Researcher_ID	Simple	Single	Primary Key
Researcher	Name	Simple	Single	Non-key
Researcher	Date_of_Birth	Simple	Single	Non-key, Stored
Researcher	Age	Simple/Derived	Single	Derived
Researcher	Email	Simple	Single	Non-key
Researcher	Phone_Number	Simple	Multi-valued	Non-key
Researcher	Address	Composite	Single	Non-key
Researcher	Street	Simple component	Single	Non-key
Researcher	City	Simple component	Single	Non-key
Researcher	State	Simple component	Single	Non-key
Researcher	PIN_Code	Simple component	Single	Non-key
Department	Department_ID	Simple	Single	PrimaryKey
Department	Department_Name	Simple	Single	Non-key
Department	Location	Simple	Single	Non-key
Research Project	Project_ID	Simple	Single	Primary Key
Research Project	Project_Title	Simple	Single	Non-key
Research Project	Start_Date	Simple	Single	Stored
Research Project	End_Date	Simple	Single	Stored
Research Project	Budget	Simple	Single	Stored
Research Project	Keyword	Simple	Multi-valued	Non-key
Student	Student_ID	Simple	Single	Primary Key
Student	Name	Simple	Single	Non-key
Student	Email	Simple	Single	Non-key
Student	Date_of_Birth	Simple	Single	Stored

Publication	Publication_ID	Simple	Single	PrimaryKey
Publication	Title	Simple	Single	Non-key
Publication	Publication_Date	Simple	Single	Stored
Publication	DOI	Simple	Single	Non-key
Publication	Publisher	Simple	Single	Non-key
Project Assistant	AssistantName	Simple	Single	Partial Key
Project Assistant	Role	Simple	Single	Non-key